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Unit II: Services & Components of operating System [10M]

- Services of OS — [Chp 1] (W-19)(2M) (S-22)(4M) (W-22)(2M)

(i) User Interface:
- All operating systems have a user interface that allows users to communicate with system. 3 types of user interfaces are available:

a) Command line Interface (CLI)
b) Batch interface
c) Graphical user interface (GUI)

(ii) I/O operations: (User)
- When a program is running, it may require input/output resources such as file or devices such as printer.
- So, OS provides a service to do I/O.

(iii) Communication: (User)
- In the system, one process may need to exchange information with another process
- Communication can be implemented via shared memory or through message passing, in which packets of information are moved between processes by OS.

(iv) Resource allocation: (OS itself)
- OS manages resource allocation to the processes.
- These resources are CPU, main memory, file storage & I/O devices.

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(v) Accounting: OS keeps track of usages of various computer resources allocated to users.

(vi) Protection & Security: (OS itself)

- When several separate processes execute concurrently, one process should not interfere with other processes or OS itself.
- Protection provides controlled access to system resources. Security is provided by user authentication such as password for accessing information.

- System Calls (W-19)(4M)
- System Call provides an interface between a running program and operating system. It allows user to access services provided by operating system.
- This system calls are procedures written using C, C++ and assembly language instructions. Each operating system has its own name for each system call.
- Each system call is associated with a number that identifies itself.
- 3 general methods are used to pass parameters between running program & OS:

(1) Pass parameters in register.

(2) Store parameters in table in memory and the table address is passed as parameter in a register.

(3) Push (store) parameters onto stack by program & pop off stack by OS.

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- System calls:

A] **Process control** - (i) Program in execution is a process. A process to be executed must be loaded in main memory, while executing it may need to wait, terminate or create & terminate child processes.

(ii) Eg - (end, abort), (load, execute), (create process, terminate process), (get process attributes, set process attributes), wait for time, (wait event, signal event), (allocate & free memory)

B] **File management** - (i) System allows us to create & delete files. For create & delete operation system call requires name of file and other attributes of file.

(ii) System access file attributes for performing operations on files & directories. Once the file is created, we can perform different operations on it.

(iii) Eg :- (create file, delete file), (open, close), (read, write, reposition), (get file attributes, set ~~device~~ file attributes) ~~(logically attach or detach devices)~~.

C] **Device management** -

(i) When a process is in running state, it requires several resources to execute. These resources include main memory, disk drives, files & so on.

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(ii) If resource is available, it is assigned to the process. Once the resource is allocated to process, process can read, write & reposition the device

(iii) eg - (request device, release device), (read, write, reposition), (get device attributes, set device attributes), (logically attach or detach devices).

D] Information maintenance -

(i) Transferring information between the user program and operating system requires system call.

(ii) System information includes displaying current date and time, the number of current user, the version number of the operating system, the amount of free memory or disk space and so on.

(iii) OS keeps information about all its processes that can be accessed with system calls.

(iv) ~~such a~~ eg: (get time or date, set time or date) (get system data, set system data) (get process, file or devices attributes) (set process, file or devices attributes)

E] Communication -

(i) Processes in the system, communicate with each other. Communication is done by using two models: message passing & shared memory.

(ii) For transferring messages, sender process connects itself to receiving process by specifying receiving process name or identity. Once the communication is over system close the connection between communicating processes.

(iii) Eg: (create, delete communication connection), (send, receive messages), (transfer status information), (attach or detach remote devices).

- Operating system Tools. (W-19) (S-22) (N-22) (6M)

1. User management
2. Security policy
3. Device management
4. Performance monitor
5. Task scheduler.

A] User management:

(i) User management includes everything from creating a user to deleting a user on your system. User management can be done in three ways on Linux system.

(ii) Command line tools include commands like useradd, userdel, usermod, passwd, etc. These are mostly user by the server administrators.

(iii) Commands:

a) useradd - With useradd commands you can add a user.

Syntax: useradd login name

Eg: useradd ~~mhssp~~ xyz

b) userdel- To delete a user account userdel command is used.
Syntax: userdel -r <username>

c) usermod- This command is used to modify the properties of an existing user.
Syntax: usermod -c <newName'> <oldName>

d) passwd - A user can set the password with command passwd. Old password has to be typed twice before entering the new one.
Syntax: passwd <username>

3) Device management:

(i) Device management is process of managing the implementation, operation and maintenance of a physical or virtual device.

(ii) All Linux device files are located in the /dev directory, which is an integral part of root(/) file system because these device files must be available to operating system during the boot process.

(iii) Eg: ls -l /dev
Above example gives the list of device file from kernel.

(iv) Udev supplies a dynamic device directory containing only the nodes for devices which are connected to the system. It creates or removes the device node files in the /dev directory.

d) netstat - Linux network & statistics monitoring tool
Displays network connections, routing tables, interface statistics, multicast memberships, etc.
netstat -tulpn

c) Performance monitor:

(i) It is very tough job for every system or network administrator to monitor and debug Linux System Performance problems every day. ~~The comm~~

(ii) The commands discussed below are some of the most fundamental commands when it comes to system analysis and debugging Linux server issues such as:

a) top - Process activity monitoring command.
~~b)~~ ~~v~~ Provides a dynamic real-time view of running system. By default, it displays most CPU-intensive tasks running on server & updates the list every five seconds. $ top

b) vmstat - Virtual memory statistics
Reports information about processes, memory, paging, traps and cpu activity $ vmstat 3

c) free - Show Linux server memory usage.
Shows total amount of free and used physical and swap memory in the system, as well as buffers used by kernel. # free

d) Security policy:

(i) For securing Linux OS there are some practices: Always keep system updated with latest releases patches, security fixes & kernel when its available.

(ii) Syntax: $ yum updates
$ yum check-update

(iii) To maintain security:
a) Use secure SSH
b) Lock & Unlock features

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c) Turn off IPv6
d) Enables Iptables (Firewall)
e) Use Strong Password policy
f) Install Antimalware/Antivirus software
g) Backup regularly

E) Task scheduler:
i) Crontab files can be used to automate backups, system maintenance and other repetitive tasks.
ii) The syntax is powerful & flexible, so you can have a task run every fifteen minutes or at specific minute on a specific day every year.

- Open system call